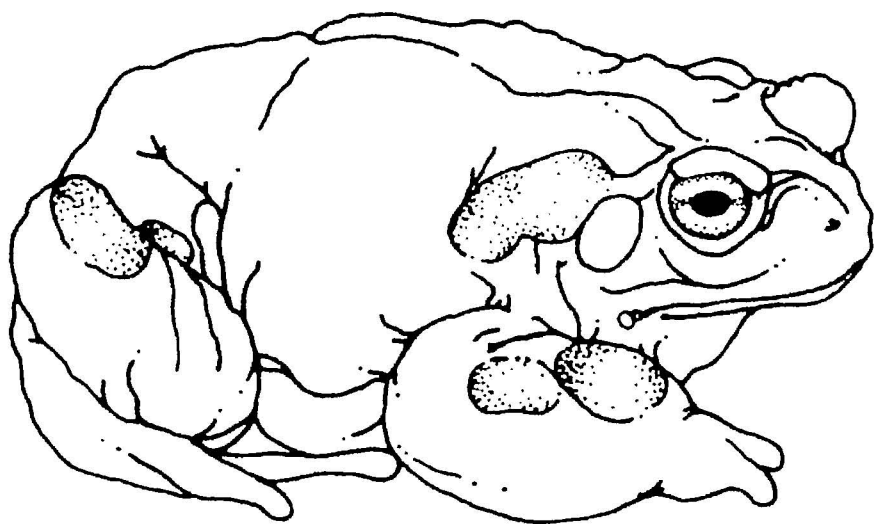


*Bufo alvarius:*  
*the*

**PSYCHEDELIC TOAD**  
*of the Sonoran Desert*



*by Ken Nelson*

*expanded and updated edition by Hamilton Morris*



*Bufo alvarius:*

*the*

**PSYCHEDELIC TOAD**

*of the Sonoran Desert*

*by Albert Most*



venom  
press

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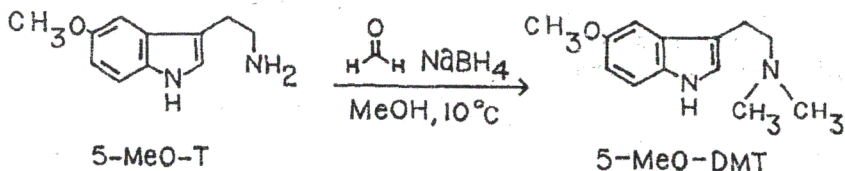
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A GREEN AND SUSTAINABLE SYNTHESIS  
OF 5-MeO-DMT

Conventional synthetic wisdom has held that sodium borohydride ( $\text{NaBH}_4$ ) is too strong of a reducing agent to be used in reductive amination procedures where an imine is formed in situ because carbonyl reduction would precede imine condensation producing an alcohol without the formation of the desired carbon-nitrogen bond. Reductive amination of tryptamines carries the additional risk of  $\beta$ -carboline side products via a pH and temperature-dependent Pictet-Spengler reaction.

A common solution to this problem has been to employ a borohydride reducing agent where one or more hydrogen atoms have been



replaced with substituents that sterically and electronically reduce reactivity, rendering it incapable of carbonyl reduction. The most common example of this is sodium cyanoborohydride (NaBH<sub>3</sub>CN) with less common use of sodium triacetoxyborohydride as well as a variety of substituted ammonia-borane hydrides. These reducing agents are considerably more expensive<sup>1</sup> than NaBH<sub>4</sub>, and, in the case of NaBH<sub>3</sub>CN, add the hazard of hydrogen cyanide production during synthesis.<sup>2</sup> Furthermore, it has been my personal experience that NaBH<sub>3</sub>CN can leave trace impurities (likely amino-cyanoborane adducts or N-cyanomethyltryptamines) in the product that smell of cyanide even after purification via column chromatography and recrystallization. The toxicity of these trace impurities is unknown, but suffice it to say they are undesirable.

As early as 1963,<sup>3</sup> chemists noted that the reactivity of NaBH<sub>4</sub> could be tempered by conducting reductive aminations at 0°C, but half a century would elapse before these techniques would be applied to tryptamine chemistry. When NaBH<sub>4</sub> was employed it was either used as part of a relatively complex multi-step alkylation or the yields were an abysmal 3% with the major product being tetrahydro- $\beta$ -carboline (TH $\beta$ C). This changed in 2014 when Yi He and Xiaolong Li demonstrated that one-pot NaBH<sub>4</sub> reduction was applicable to the synthesis of rizatriptan, a ring-substituted analog of DMT, provided that the reaction was conducted under non-acidic conditions and kept below 20°C. This work inspired several chemists in both academic and clandestine

laboratories to reproduce and extend the scope of the technique. At the time of writing,  $\text{NaBH}_4$  reductive amination has been successfully used in the synthesis of DMT, 5-MeO-DMT, DPT, NsBT and other tryptamines from a gram to multi-kilogram scale. It is my opinion that the following process represents the cheapest, easiest, safest, and most environmentally friendly method for non-enzymatic production of 5-MeO-DMT.

### 5-METHOXYTRYPTAMINE

5-methoxytryptamine is prohibitively expensive via most domestic chemical supply companies, but can be purchased inexpensively from China. The freebase is a light brown powder that smells pleasantly of grape and melts between  $121.0$ – $123.0^\circ\text{C}$ . It can be purified via recrystallization in 95% ethanol. Purchasing 5-methoxytryptamine is cheaper, faster, and less wasteful than producing it oneself, but it can be prepared from melatonin, a widely available sleep aid, in the following manner.

30 g melatonin (0.12915 moles) is suspended in 150 mL isoamyl alcohol containing 30 mL MeOH to facilitate dissolution. With vigorous stirring, 2.14 g sodium dithionite (0.01229 moles) is added followed by 16 g freshly crushed NaOH pellets (0.40003 moles). Care must be taken to ensure the NaOH does not form a solid cake that prevents stirring. The reaction is then heated to reflux under a blanket of inert gas. The initially transparent yellow solution will acquire a tan color and thicken. After 2.5 h reflux with monitoring the deacetylation is complete as indicated by GC-MS. The reaction is removed from heat, cooled to RT and added to 300 mL distilled  $\text{H}_2\text{O}$ . The aqueous phase is extracted with 4 x 60 mL toluene. The organic phases are pooled and washed with  $\text{H}_2\text{O}$  (4 x 100 mL) and brine (3 x 100 mL). The organic phase is then extracted with 3N aqueous HCl (3 x 120 mL). The acidic

aqueous extracts are pooled, made basic with NaOH pellets and extracted with ethyl acetate (3 x 100 mL). The pooled organic extracts are washed with saline (1 x 10 mL), dried with anhydrous sodium sulfate, and evaporated under reduced pressure to yield an amber oil which spontaneously sets to a tan crystalline solid. The solids are recrystallized by dissolving in 50 mL 95% ethanol at -20°C overnight. The resulting crystals are collected by decanting the solvent, washing with a small amount of ice-cold ethanol, and drying in an oven at 60°C. The tan solids are crystallized once more as above to give 11.4 g large transparent amber crystals of 5-methoxytryptamine. Additional crops can be obtained but require further purification.

#### 5-METHOXY-N,N-DIMETHYLTRYPTAMINE

The following procedure is a synthesis of personal experience and reports I have collected from other chemists. Although this process is a refinement of many similar procedures, it is by no means optimized. I will describe some problems I have encountered and how best to avoid them. The potential synthesist should recognize that the ideal conditions are dependent on reaction scale and availability of reagents and equipment. Some chemists will have access to a mechanically stirred three-neck borosilicate reaction kettle that is jacketed and chilled with a recirculating reservoir of -45°C aqueous ethylene glycol; other chemists will be limited to a stainless steel soup pot and a Rubbermaid tote filled with salted ice - plan accordingly.

100 g 5-methoxytryptamine freebase (0.52564 moles) is poured into a 10 L, three-neck reaction kettle nested inside a 3.5 L ice bath. With magnetic stirring, the tryptamine is dissolved in 4.5 L of MeOH forming a translucent brown solution. The bath is filled with pellets of dry ice, followed by MeOH and

the 5-methoxytryptamine solution is allowed to cool until it has reached 0°C.

100 g  $\text{NaBH}_4$  (2.643 moles)<sup>4</sup> is dissolved in 250 mL (10.6 M) of a 3% KOH solution.<sup>5</sup> This solution should be cooled in an ice bath to at least 15°C before being transferred into an addition funnel with a pressure-equalization arm.<sup>6</sup> Charge a second addition funnel with 400 mL (5.37 moles) 37% formaldehyde solution<sup>7</sup> and begin a dropwise addition,<sup>8</sup> carefully monitoring the temperature so that it stays between -5 and 15°C.<sup>9</sup>

Once the formaldehyde and  $\text{NaBH}_4$  have been added, it is advisable to begin monitoring the reaction via TLC with a mobile phase of 4:1 MeOH:EtOAc.<sup>10</sup> The rank order of R<sub>f</sub> values should be as follows: 5-MeO-T<5-MeO-NMT<5-MeO-DMT<6-MeO-TH $\beta$ C. I have only encountered prominent spots for the starting material and the product.

Once TLC or mass spectrometry have demonstrated that reductive amination is complete, the solution can be allowed to warm to room temperature and borate salts will begin settling to the bottom of the vessel. At this point the supernatant can be decanted or, if one has access to a sufficiently large Buchner funnel, the entire reaction mixture can be subjected to vacuum filtration and the solvent can be stripped from the filtrate using a rotary evaporator. If a rotary evaporator is unavailable the filtered solution can be boiled with stirring, but if the borate salts are not filtered off before the boiling begins they can coat the bottom of the vessel and cause uneven heating, leading to decomposition of product and breakage of glassware.

MeOH (BP 64-65°C) will boil off before water and as the MeOH concentration decreases, the 5-MeO-DMT freebase oil will precipitate from the solution and rise to the surface as

a dark brown oil. When the boiling point of the solution reaches  $93^{\circ}\text{C}$ , the concentration of MeOH is less than 10% and the solution is sufficiently polar to allow extraction with an organic solvent.

At this point, the scale of the reaction and availability of equipment for purification will strongly influence how best to proceed. The 5-MeO-DMT freebase can be extracted with ethyl acetate then dried with sodium sulfate, and the solvent can be stripped from the base with a rotary evaporator. The crude base may crystallize spontaneously after sitting at room temperature and the crystals can be purified by dissolution in boiling hexanes or heptane and left to cool to room temperature, at which point the solution should be transferred into a refrigerator and allowed to crystallize overnight. The recrystallization can be repeated as many times as necessary to achieve a product of satisfactory purity. The crude product can also be filtered through a plug of silica gel to remove colored impurities and then subjected to additional purification.

Due to the vast quantities of organic solvent required for recrystallization, larger scale reactions may benefit from purification via short path vacuum distillation. The crude 5-MeO-DMT base is allowed to dry so that any residual water or methanol are no longer present, then transferred into a short path distillation apparatus, ideally with a heating mantle and vacuum capable of pulling at least 1.0 mm/Hg. 5-MeO-DMT freebase boils at  $160\text{--}170^{\circ}\text{C}$  at 0.6 mm/Hg. Care should be taken to allow the distillate to cool to room temperature before breaking the vacuum to avoid rapid thermal decomposition. The distillate may spontaneously crystallize on cooling and can be recrystallized to yield 5-MeO-DMT of exceptional purity with a MP of  $145\text{--}146^{\circ}\text{C}$ .

## ENDNOTES

1.  $\text{NaBH}_4$  is typically less than half the cost of other borohydride reducing agents and is commercially available at higher purities. At the time of writing, one kilogram of 98%  $\text{NaBH}_4$  can be ordered from China for \$220.

2. If  $\text{NaBH}_3\text{CN}$  were employed in a reductive amination at the scale described in this book it would have the theoretical capacity to evolve more than 22 g of hydrogen cyanide, a fatal dose many times over. In practice, proper ventilation and control of reaction conditions make  $\text{NaBH}_3\text{CN}$  a reasonably safe reagent, but alternatives should be pursued when they are available.

3. Karl A. Schellenberg, 1963, The Synthesis of Secondary and Tertiary Amines by Borohydride Reduction.

4. Every mole of  $\text{NaBH}_4$  has the theoretical ability to reduce four moles of imine, but the  $\text{NaBH}_4$  is also reacting with the solvent system and likely begins to lose reductivity after donation of the first hydride, for this reason it is used in stoichiometric excess.

5. It is better to add a saturated  $\text{NaBH}_4$  solution with a small amount of suspended, undissolved  $\text{NaBH}_4$  than it is to introduce the additional water required to dissolve it. Some chemists have successfully carried out this procedure adding the  $\text{NaBH}_4$  to the reaction mix as a solid powder. Although it can be done, this increases the likelihood of thermal spikes producing  $\beta$ -carboline side products and a small amount of water increases the solubility of the  $\text{NaBH}_4$  in cold MeOH. Conversely, too much water will slow the evaporative workup and cause tryptamine bases to aggressively precipitate forming an intractable brown tar that will trap a magnetic stir bar and cause considerable frustration if an overhead stirrer and additional MeOH are not available to facilitate redissolution. A carefully controlled ratio of water to MeOH preserves the amphiphilicity of the solvent system allowing both tryptamine bases (non-polar) and  $\text{NaBH}_4$  (polar) to remain in solution. A rule of thumb is to never let the water concentration of the reaction mix exceed 10%.

6. 40%  $\text{NaBH}_4$  should represent a near saturated solution at  $12.5^\circ\text{C}$ . Although  $\text{KOH}$  increases the stability of aqueous  $\text{NaBH}_4$ , its gradual decomposition will be evident from the continuous evolution of hydrogen gas. This reaction should not be carried out in a closed vessel because the pressure will build and, at the very least, pop a joint or septum.

7. Typical commercial formaldehyde solutions are 48-53% water with 10-15%  $\text{MeOH}$ , which is used as a stabilizer to prevent formaldehyde's polymerization into paraformaldehyde.

8. Many chemists have used a simultaneous dropwise addition of the formaldehyde and  $\text{NaBH}_4$  solutions. A mechanistic argument can be made for alternating additions, but the reaction will rapidly proceed to completion as long as there is not a tremendous excess of  $\text{NaBH}_4$ , which reduces the formaldehyde before it has time to condense with the amine.

9. 100 mL of 37% formaldehyde contains 1.34 moles and 100 mL of a 40%  $\text{NaBH}_4$  solution contains 1.57 moles, thus the  $\text{NaBH}_4$  solution is more concentrated. Accordingly, the rate of formaldehyde addition should be faster than the rate of  $\text{NaBH}_4$  addition. This reductive amination occurs rapidly and the addition can be carried out quickly on a small scale, but a larger scale necessitates an especially cautious addition because both imine formation and  $\text{NaBH}_4$  reduction are exothermic and it is crucial for the reaction to be kept between  $-10^\circ\text{C}$  and  $15^\circ\text{C}$ . Anything over  $20^\circ\text{C}$  risks a Pictet-Spengler side reaction that will produce pinoline (6-MeO-TH $\beta$ C) and N-methyl-pinoline (6-MeO-2-Me-TH $\beta$ C).

10. This mobile phase produces less than satisfactory separation, but it has the advantage of reusing solvents already present for the reaction. An improved solvent system is pure ethanol or methanol with 1% triethylamine.



Specialized multi-cellular glands concentrated on the neck and limbs of B. alvarius produce a viscous milky-white venom that contains large amounts of the potent hallucinogen 5-MeO-DMT. When vaporized by heat and taken into the lungs in the form of smoke, this indole-based alkaloid produces an incredibly intense psychedelic experience of extremely short duration. There is no hangover or harmful effect. On the contrary, a pleasant psychedelic afterglow appears quite regularly after smoking the venom of B. alvarius, the Psychedelic Toad of the Sonoran Desert.

In this expanded and updated edition of the 1983 ethnoherpetological classic, the story of how Ken Nelson (pseud. Albert Most) discovered and smoked the psychedelic venom of B. alvarius is told for the first time. Additionally, a green and sustainable 5-MeO-DMT synthesis is presented to protect natural toad populations and accommodate the growing international demand for this extraordinary psychedelic.

